

Agri-Shokti: A Multimodal AI-Driven Precision Agriculture Framework for Smallholder Farmers in Bangladesh

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Abstract—Smallholder farmers in developing regions like Bangladesh face significant productivity challenges due to a lack of real-time, localized, and scientifically grounded agricultural advice. While satellite remote sensing offers high-level insights, it often fails to translate into actionable steps for individual farmers due to low literacy rates and the disconnect between orbital data and ground-truth conditions. This paper presents **Agri-Shokti**, a comprehensive Decision Support System (DSS) that bridges this gap. The system employs a novel **Satellite-to-Ground-Truth Loop**, triggering localized alerts based on Sentinel-2 NDVI anomalies which farmers verify via smartphone photography. We propose a four-stage reasoning pipeline: (1) **Contextual Sensing** using satellite and weather data, (2) **Adaptive Reasoning** via Multimodal Large Language Models (Gemini 2.5 Flash), (3) **Expert Validation** using Retrieval-Augmented Generation (RAG) against national agricultural standards (BARC/FAO) to minimize hallucination, and (4) **Hyper-Localized Delivery** through a Bangla voice-first interface. Experimental prototype results demonstrate a 92% diagnostic accuracy and a system latency of under 4 seconds, offering a scalable, low-bandwidth solution for enhancing crop resilience and food security in the Global South.

Keywords—Precision Agriculture, Multimodal AI, Retrieval-Augmented Generation (RAG), Satellite Remote Sensing, Smallholder Farming, ICT4D.

1. Introduction

Agriculture is the economic backbone of Bangladesh, employing roughly 40% of the workforce and contributing significantly to GDP. However, the sector is plagued by low productivity, climate change vulnerability, and informational inefficiencies. Traditional extension services are often understaffed, with ratios as low as one extension agent for every 3,000 farmers [1]. Consequently, farmers rely on heuristic knowledge or untrained local input dealers, leading to the overuse of fertilizers, improper pest management, and yield losses.

To address these limitations, we introduce **Agri-Shokti** (meaning "Agricultural Power"), a web-based AI platform designed specifically for the Bangladeshi context. Agri-Shokti lowers the barrier to entry for precision agriculture by utilizing ubiquitous smartphone technology to deliver expert-level, verified agronomic advice.

2. Related Work

A. IoT and AI in Smart Farming

Recent literature highlights the transformative potential of Internet of Things (IoT) and Artificial Intelligence (AI) in agriculture. Studies by *Dharmaraj et al. (2021)* demonstrate the efficacy of CNNs in plant disease detection. However, most existing models are trained on controlled datasets (e.g., PlantVillage) and struggle with the "in-the-wild" conditions typical of Bangladeshi farms (poor lighting, background noise).

B. Remote Sensing Limitations

Remote sensing using Sentinel-2 and Landsat data is well-established for large-scale crop monitoring [3]. Indices like NDVI (Normalized Difference Vegetation Index) correlate well with crop vigor. Yet, for smallholders with fragmented land plots (<0.5 acres), satellite resolution often blends mixed pixels, necessitating ground-truth verification which remains a logistical bottleneck.

3. System Architecture

The Agri-Shokti architecture mimics a node-based automation workflow, consisting of four distinct stages as shown in Fig. 1.

```
graph TD
    User[Farmer (User)] -->|Uploads Image| App[Agri-Shokti App]
    App -->|Edge Function| Logic[Reasoning Engine]
    subgraph Cloud_Backend [Cloud Backend]
        Logic -->|Vision Analysis| Gemini[Gemini 2.5 Flash]
        Logic -->|RAG Query| VectorDB[(Supabase Vector Store)]
        VectorDB -->|Contextual Protocol| Logic
    end
    Gemini -->|Disease ID| Logic
    Logic -->|Verified Advice| TTS[Text-to-Speech Engine]
    TTS -->|Voice Output| User
    style Logic fill:#f9f,stroke:#333
    style Gemini fill:#bbf,stroke:#333
    style VectorDB fill:#cfc,stroke:#333
```

Fig. 1. High-level System Architecture of Agri-Shokti showing the interaction between the User, App, and the AI/RAG Backend.

A. Stage 1: Contextual Sensing ("The Eyes")

- **Remote Sensing:** Fetches Sentinel-2 data via **AgroMonitoring API**.
- **Weather:** **NASA POWER** integration for evapotranspiration and solar data.
- **User Input:** Smartphone camera and geolocation.

B. Stage 2: Adaptive Reasoning ("The Brain")

- **Model:** **Google Gemini 2.5 Flash** (Multimodal).
- **Function:** Processes visuals (leaf health) and context (weather/season) to form a hypothesis.

C. Stage 3: Expert Validation ("The Library")

- **Mechanism:** Retrieval-Augmented Generation (RAG).
- **Database:** Supabase pgvector stores embeddings of BARI/FAO guidelines.
- **Logic:** If AI confidence < Threshold OR advice contradictions vector retrieval, fallback to retrieved content.

4. Methodology

The system methodology follows a standard inference cycle:

1. **Data Acquisition:** User captures an image; system simultaneously fetches metadata (Lat/Long, Weather, Date).
2. **Preprocessing:** Image is resized/compressed; metadata is formatted into a prompt context.
3. **Inference:**
 - *Vision Pass:* Detect potential pathogens.
 - *Logic Pass:* Select treatment based on pathogen + crop stage.
4. **Verification:** Semantic search against BARI database to validate chemical dosages.
5. **Response Generation:** Construct JSON response with 'speakable' text and UI cards.

5. Experimental Results

To validate the efficacy of Agri-Shokti, we conducted a comparative study of diagnostic accuracy and system latency.

A. Diagnostic Accuracy

We compared the disease detection accuracy of our RAG-enhanced method against a standard MobileNetV2 CNN and a zero-shot Gemini Vision prompt (without RAG). The setup involved 50 test images of common diseases (Rice Blast, Bacterial Blight, Brown Spot).



Figure 2: Accuracy Comparison *Fig. 2. Diagnostic Accuracy of various approaches. Agri-Shokti (92%) outperforms standard CNNs and Zero-shot LLMs.*

As shown in Fig. 2, the **Agri-Shokti (RAG)** approach achieved **92% accuracy**. This significant improvement over the Zero-shot baseline (78%) is attributed to the RAG layer, which grounds the LLM's visual analysis in specific local agricultural protocols, reducing "hallucinations" of non-native diseases.

B. System Latency

Latency is a critical factor for rural usability where network connectivity is often unstable (2G/3G). We measured the average round-trip time for a full diagnostic cycle (Upload -> Analysis -> Voice Response) across different network conditions.



Figure 3: System Latency *Fig. 3. Average System Latency across different network profiles.*

Fig. 3 illustrates that even on 3G networks, the system responds in approximately **8 seconds**, which is acceptable for an asynchronous advisory tool. In **Offline Mode**, cached advice is delivered instantly (0.5s).

6. Conclusion

Agri-Shokti demonstrates that advanced AI and remote sensing can be effectively "downscaled" to serve smallholder farmers. By combining the "eyes" of NASA satellites with the "brain" of multimodal AI and the "wisdom" of national

research institutes, we created a system that is both cutting-edge and contextually grounded. Future work will focus on integrating Synthetic Aperture Radar (SAR) data to overcome cloud cover limitations during the monsoon season.

References

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